

How to Put a Heavier Higgs on the Lattice

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Abstract

Lattice work, exploring the Higgs mass triviality bound, seems to indicate that a strongly interacting scalar sector in the minimal standard model cannot exist while low energy QCD phenomenology seems to indicate that it could. We attack this puzzle using the $1/N$ expansion and discover a simple criterion for selecting a lattice action that is more likely to produce a heavy Higgs particle. Our large N calculation suggests that the Higgs mass bound might be around 850 GeV , which is about 30% higher than previously obtained.

1 Introduction

Lattice work, exploring the Higgs mass triviality bound, seems to indicate that a strongly interacting scalar sector in the minimal standard model cannot exist while low energy QCD phenomenology seems to indicate that it could. We attack this puzzle using the $1/N$ expansion and discover a simple criterion for selecting a lattice action that is more likely to produce a heavy Higgs particle. Our large N calculation suggests that the Higgs mass bound might be around 850 GeV , which is about 30% higher than previously obtained.

2 Method

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$$L(\theta) = \sum_{i=1}^n \ell(x_i, \theta) \tag{1}$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

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